

26 - Reaction Kinetics

Rate of reaction:

- the rate of change in the concentration of a particular reactant
- affected by temperature, concentration, catalyst etc.

$$\Rightarrow \frac{\text{change in concentration}}{\text{time}}$$

Rate equation Can only be found from experimental data

* for a general equation $A + B \rightarrow C$

$$\text{rate} = k[A]^m[B]^n$$

$[A]$ & $[B]$ are conc. of reactants & m/n are their orders respectively. k is the rate constant (proportionality constant)

* not all reactants have to show in the rate equation

* rate equation may include catalysts



① find how $[\text{H}_2]$ affects rate by varying $[\text{H}_2]$ while keeping $[\text{NO}]$ constant

↳ from experimental data, the rate is proportional to $[\text{H}_2]$

$$\text{rate} = k_1[\text{H}_2]$$

② find how $[\text{NO}]$ affects rate by varying $[\text{NO}]$ while keeping $[\text{H}_2]$ constant

↳ from experimental data, the rate is proportional to the square of $[\text{NO}]$

$$\text{rate} = k_2[\text{NO}]^2$$

⇒ thus rate equation is: $\text{rate} = k[\text{H}_2][\text{NO}]^2$

units of k : DEPENDS

e.g. $\text{rate} = k[\text{H}_2][\text{I}_2]$

$$k = \frac{\text{rate}}{[\text{H}_2][\text{I}_2]}$$

$$k = \frac{\text{mol dm}^{-3} \text{s}^{-1}}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$$

$$k = \text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$$

effect of temperature on rate constant k :

- higher temperature
- greater proportion of particles have energy $\geq E_a$
- higher frequency of successful collisions
- rate of reaction increases
- rate constant also increases

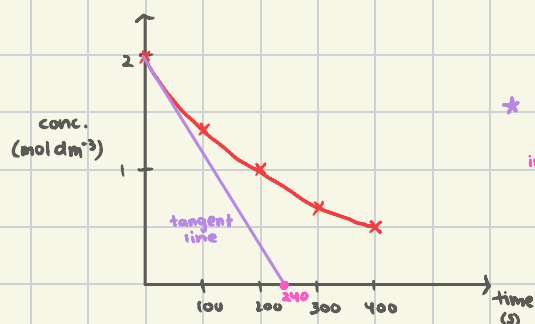
example Q: a student investigates the concentration of reactant X

The student starts with 2.00 mol dm^{-3} of X
 & measures the conc. of X every 100 sec.

→ Plot a graph & determine initial rate of reaction

This is the results:

time (s)	$[\text{H}_2\text{O}_2] (\text{mol dm}^{-3})$
0	2.00
100	1.40
200	1.00
300	0.60
400	0.50



* initial rate of reaction:

$$\begin{aligned} \text{initial rate} &= \frac{\Delta \text{conc.}}{\Delta \text{time}} \\ &= \frac{2.00 - 0.00}{240 - 0} \\ &= 8.3 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1} \end{aligned}$$

Order of reaction:

* the order of reaction w.r.t a certain reactant is the power to which it is raised to in the rate equation

* can be 0, 1, or 2

e.g rate = $k[\text{H}_2][\text{NO}]^2$

first order w.r.t $[\text{H}_2]$

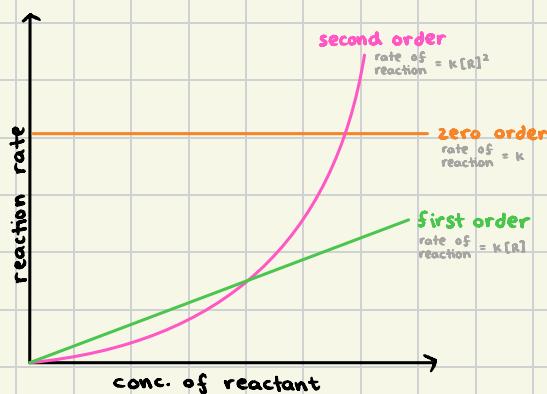
second order w.r.t $[\text{NO}]$

overall order = $1 + 2 = 3$

How to identify order of reaction:

- 1 plot a graph of reaction rate against concentration of reactant
- 2 plot a graph of concentration of reactant
- 3 deduce successive half-lives from graphs of concentration against time

graph of reaction rate against concentration



zero order: e.g $2\text{NH}_3 \longrightarrow \text{N}_2 + 3\text{H}_2$
 rate = $k[\text{NH}_3]^0 = k$

reaction rate is NOT affected by concentration

first order: e.g $2\text{N}_2\text{O} \longrightarrow 2\text{N}_2 + \text{O}_2$
 rate = $k[\text{N}_2\text{O}]^1 = k[\text{N}_2\text{O}]$

reaction rate is directly proportional to $[\text{N}_2\text{O}]$

second order: e.g $\text{NO}_2 + \text{CO} \longrightarrow \text{NO} + \text{CO}_2$
 rate = $k[\text{NO}_2]^2$

reaction rate is directly proportional to the square of $[\text{NO}_2]$

doubling a reactant with first order would double rate, while doubling a second order reactant would quadruple rate

plot a graph of concentration of reactant



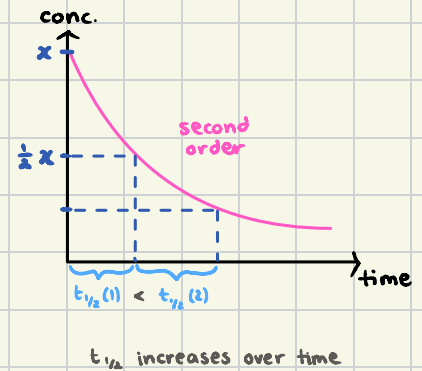
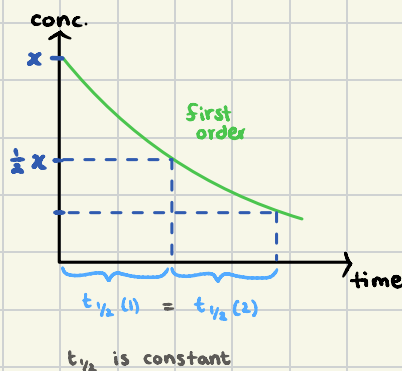
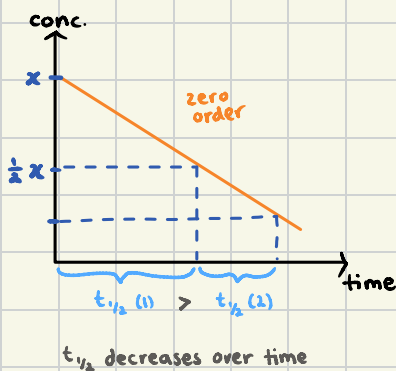
⇒ zero order reaction is a straight line

⇒ first & second order reactions are curves & we can distinguish b/w them by determining the successive half-lives of the reaction

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deduce successive half-lives from graphs of concentration against time

Half-life, $t_{1/2}$: time taken for the concentration of a limiting reactant to fall to half of its initial value



calculating k from half-life: only need to know for first-order reactions



use the expression $\star t_{1/2} = \frac{\ln 2}{k} \star$

units: $k = \frac{\text{mol dm}^{-3} \text{ s}^{-1}}{\text{mol dm}^{-3}}$
 $k = \text{s}^{-1}$

$$k = \frac{0.693}{t_{1/2}} \longrightarrow k = \frac{0.693}{17 \times 60 \text{ seconds}} \longrightarrow k = 6.79 \times 10^{-4} \text{ s}^{-1}$$

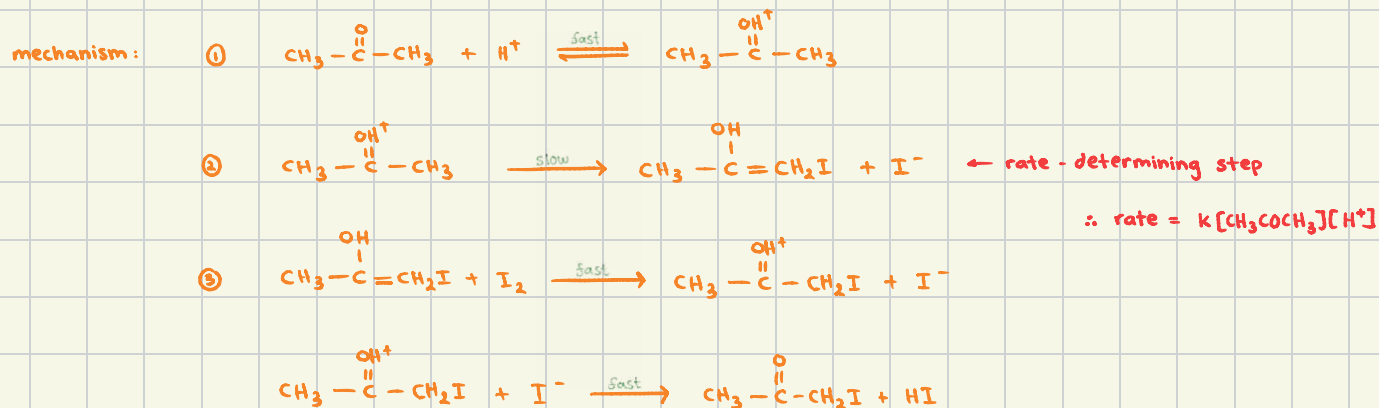
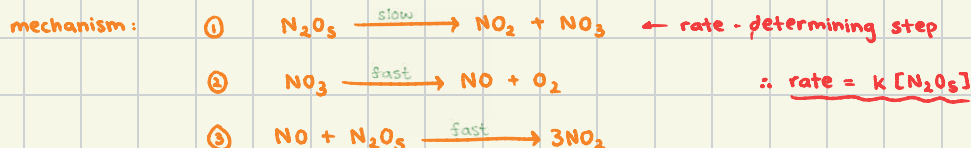
Rate - determining step

⇒ in a reaction mechanism, the steps don't all take place at the same time

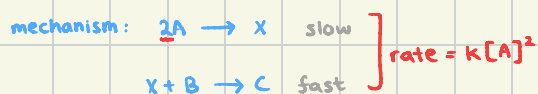
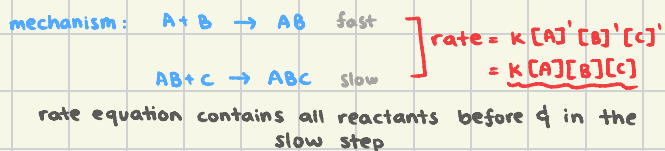
⇒ the slowest step is called the rate-determining step

⇒ only the reactants involved in or before the rate determining step appear in the rate equation

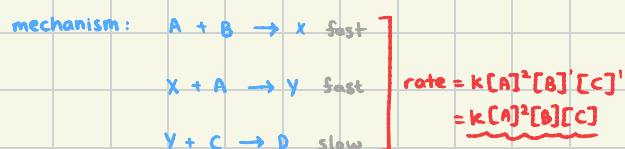
⇒ the order of a reactant tells us how many molecules of the reactant were involved in the mechanism til the slow step



Predicting the order of a reaction from reaction mechanisms



the order of a species tells us how many molecules of the reactant were involved in the mechanism til the slow step



whats an intermediate?

↳ e.g overall: $A + D \rightarrow E$

mechanism: $A + \textcircled{B} \rightarrow \underline{C}$ — made in ① & used up in ② so C is the intermediate

$C + D \rightarrow E + \textcircled{B}$ — regenerated so B acts as a catalyst

How to determine the rate-determining step from given reaction mechanisms:

↳ slow step would be the step where the 'total' reactants match the rate equation

e.g rate = $k[A]$

step 1: $\underline{A} \rightarrow B$ slow ← the 'total' reactants matches the rate equation

step 2: $B + C \rightarrow D$

rate = $k[A][B]$

step 1: $\underline{A} + A \rightarrow A_2$

step 2: $A_2 + \underline{B} \rightarrow C + A$ slow ← the 'total' reactants matches the rate equation

rate = $k[X]^2$ ← 2 molecules of X are used until slow step

step 1: $\underline{X} + Y \rightarrow XY$

step 2: $XY + \underline{X} \rightarrow X_2 + Y$ slow ← the 'total' reactants matches the rate equation

Real eg.

$NO_2 + CO \rightarrow NO + CO_2$

rate = $k[NO_2]^2$

step 1: $\underline{2NO_2} \rightarrow NO_3 + NO$ slow ← the 'total' reactants matches the rate equation

step 2: $NO_3 + CO \rightarrow NO_2 + CO_2$

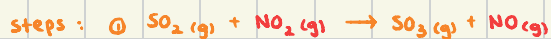
Catalyst:

↳ substances that increase the rate of reaction by offering an alternative reaction pathway with a lower E_a

Homogenous catalysts

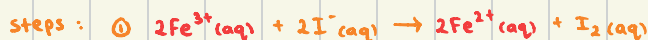
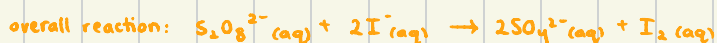
- when the catalyst is in the same phase as the reaction mixture
- are used up in one step & reformed in a later step (regenerated)

example 1: Atmospheric NO_x catalysing oxidation of SO_2



★ NO_x acts as homogenous catalyst, as they are in the same phase as reaction mixture & regenerated

example 2: iodine-peroxodisulfate reaction



reduction of Fe^{3+} to Fe^{2+} by I^-



oxidation of Fe^{2+} back to Fe^{3+} by $\text{S}_2\text{O}_8^{2-}$

★ Fe^{2+} & Fe^{3+} act as homogenous catalysts, as they are in the same phase as reaction mixture & regenerated

Heterogenous catalysts

- when the catalyst is in a different phase to the reaction mixture
- works in three stages:
 - ↳ adsorption
 - ↳ bond-weakening
 - ↳ desorption

→ reactant diffuses to surface of catalyst

physical adsorption → weak forces link the reactant to the catalyst

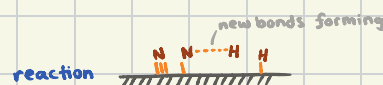
chemical adsorption → stronger bonds form b/w reactant & catalyst

→ reactants covalent bonds weaken

→ activation energy is lowered

desorption → when reaction is complete on catalysts surface, bonds b/w product & catalyst weaken so much that product breaks away

the bonds formed are strong enough to weaken the covalent bonds of the reactants & weak enough to break & allow product to leave surface



example 1: Haber process

overall reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

- ① N_2 & H_2 diffuse to surface of iron
- ② reactant molecules are adsorbed onto the surface of iron & their covalent bonds weaken
- ③ adsorbed nitrogen & hydrogen atoms react on the surface of the iron to form NH_3 (ammonia)
- ④ bonds b/w ammonia & surface of iron weaken & eventually break (desorption)
- ⑤ ammonia diffuses away from surface of iron

example 2: Pd, Pt or Rh in car catalytic converters

overall reaction: $2\text{NO}(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{N}_2(\text{g}) + \text{CO}_2(\text{g})$ & $2\text{NO}_2(\text{g}) + 4\text{CO}(\text{g}) \rightarrow \text{N}_2(\text{g}) + 4\text{CO}_2(\text{g})$

- ① nitrogen oxides & carbon monoxide diffuse to the surface of Pd, Pt or Rh
- ② reactant molecules are adsorbed onto the surface of catalyst & their covalent bonds weaken
- ③ adsorbed nitrogen oxide & carbon monoxide react on the surface of the iron to form N_2 & CO_2
- ④ bonds b/w N_2 / CO_2 & surface of catalyst weaken & eventually break (desorption)
- ⑤ products diffuse away from surface of catalyst

Summary of mechanism of a heterogeneous catalyst:

- 1) adsorption of reactants onto the catalyst surface
- 2) weakening of covalent bonds within the reactants (E_a lowered)
- 3) formation of new bonds b/w the reactants to form product (reaction)
- 4) desorption of products from surface of catalyst